

Detection and Classification of Small-sized UAVs and Birds in Sparse LiDAR Point Cloud

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Motivation and Problem Statement

- Consumer drones pose safety and privacy risks for critical infrastructure
- Existing RADAR and camera-based methods have limitations for 3D detection
- Rosette scanning LiDAR provides sparse point clouds, making detection challenging
- Key challenge: Distinguishing UAVs from birds and filtering false positives
- Goal: Real-time detection and classification using only LiDAR data

Related Work

- Prior work uses octomap-based static background subtraction
- Camera and RADAR methods lack 3D spatial accuracy or require expensive hardware
- Existing methods struggle to distinguish UAVs from birds in dynamic scenes
- Our approach: LiDAR-only detection and classification up to 100 m

Proposed Solution Pipeline

- Raw point cloud input from rosette scanning LiDAR
- Semantic segmentation removes static and dynamic background
- Foreground points clustered and preprocessed
- Modified PointNet classifies objects as drone or bird

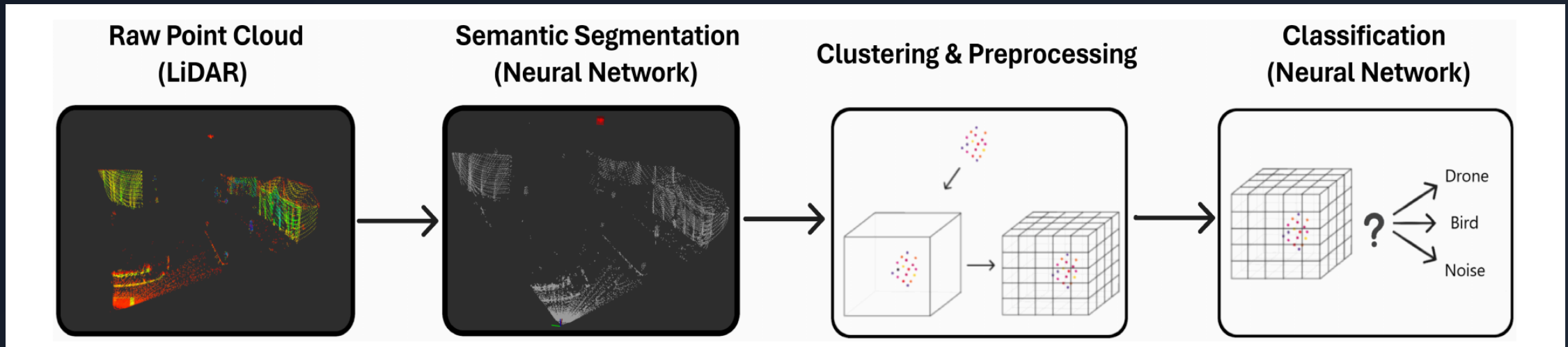


Fig. 1. Outline of the proposed solution: The raw point cloud stream, that is provided by a rosette scanning LiDAR serves as input for an attention-based neural network that removes both the static and dynamic background elements from it. The remaining foreground points then undergo a clustering and a unique preprocessing step in order to be classified by a slightly modified PointNet network.

Semantic Segmentation Network

- Spatial transformer (T-Net) aligns point clouds to a uniform space
- Multi-head attention captures global context between points
- Directly uses points as tokens, skipping voxelization
- MLP assigns background or foreground class to each point

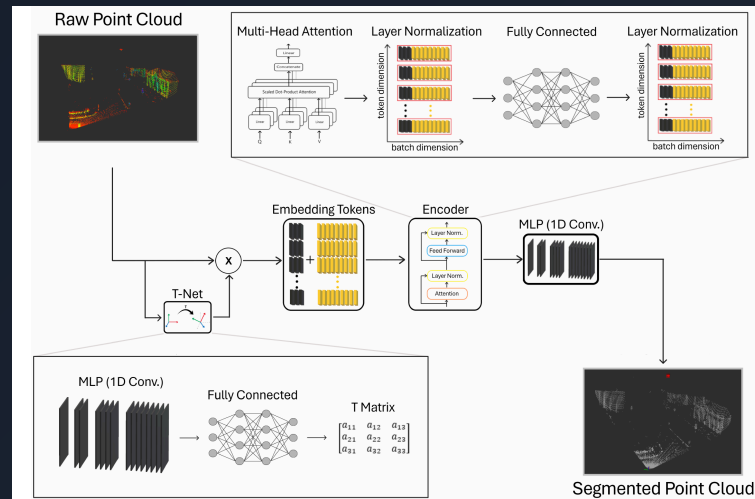


Fig. 2. Overview of the semantic segmentation network: the point cloud is transformed into a transformation invariant space, where each of the points influence is calculated with relevance to all the other points through the Multi-Head attention [17] mechanism and finally, all of them are assigned with a class determined by an MLP.

Preprocessing and Classification

- Background points removed; foreground points clustered
- Clusters with fewer than 3 points discarded as noise
- Nearest Neighbour Search recovers missed foreground points
- Voxel grid ensures uniform input for classification
- Modified PointNet classifies clusters as drone, bird, or unknown

Component-wise Inference Speed

- All pipeline components achieve real-time speeds
- Mean processing time per frame: segmentation 4.29 ms, clustering 1.65 ms, preprocessing 4.11 ms, classification 6.44 ms
- System operates at >10 FPS, suitable for real-time applications

	Segmentation	Clustering	Preprocessing	Classification
Worst-case	6.96 ms	8.86 ms	16.82 ms	8.14 ms
Deviation	0.49 ms	1.41 ms	2.46 ms	0.39 ms
Mean	4.29 ms	1.65 ms	4.11 ms	6.44 ms

Table 1 COMPONENT-WISE PERFORMANCE REGARDING INFERENCE SPEED

Background Removal Speed Comparison

- Dynamic background removal (ours) is significantly faster than static octomap-based method
- Mean runtime: 4.29 ms (dynamic) vs 16.49 ms (static)
- Lower deviation and improved handling of dynamic environments

	Dynamic (ours)	Static [5]
Worst-case	6.96 ms	57.27 ms
Deviation	0.49 ms	9.65 ms
Mean	4.29 ms	16.49 ms

Table 2 STAT1C AND DYNAM1C BACKGROUND REMOVAL SPEED COMPAR1SON

Semantic Segmentation Performance

- Balanced F-Measure (F1): 0.78 on test dataset
- Low false positives; most objects are detected
- False negatives mainly appear as small noisy clusters

Precision	Recall	F-Measure
0.7634	0.7949	0.7787

Table 3 PERFORMANCE EVALUATION OF THE SEMANTIC SEGMENTATION NETWORK

Classification Network Results

- Classification evaluated as part of full pipeline
- Test dataset balanced between drones and birds
- Overall accuracy: 81.5%

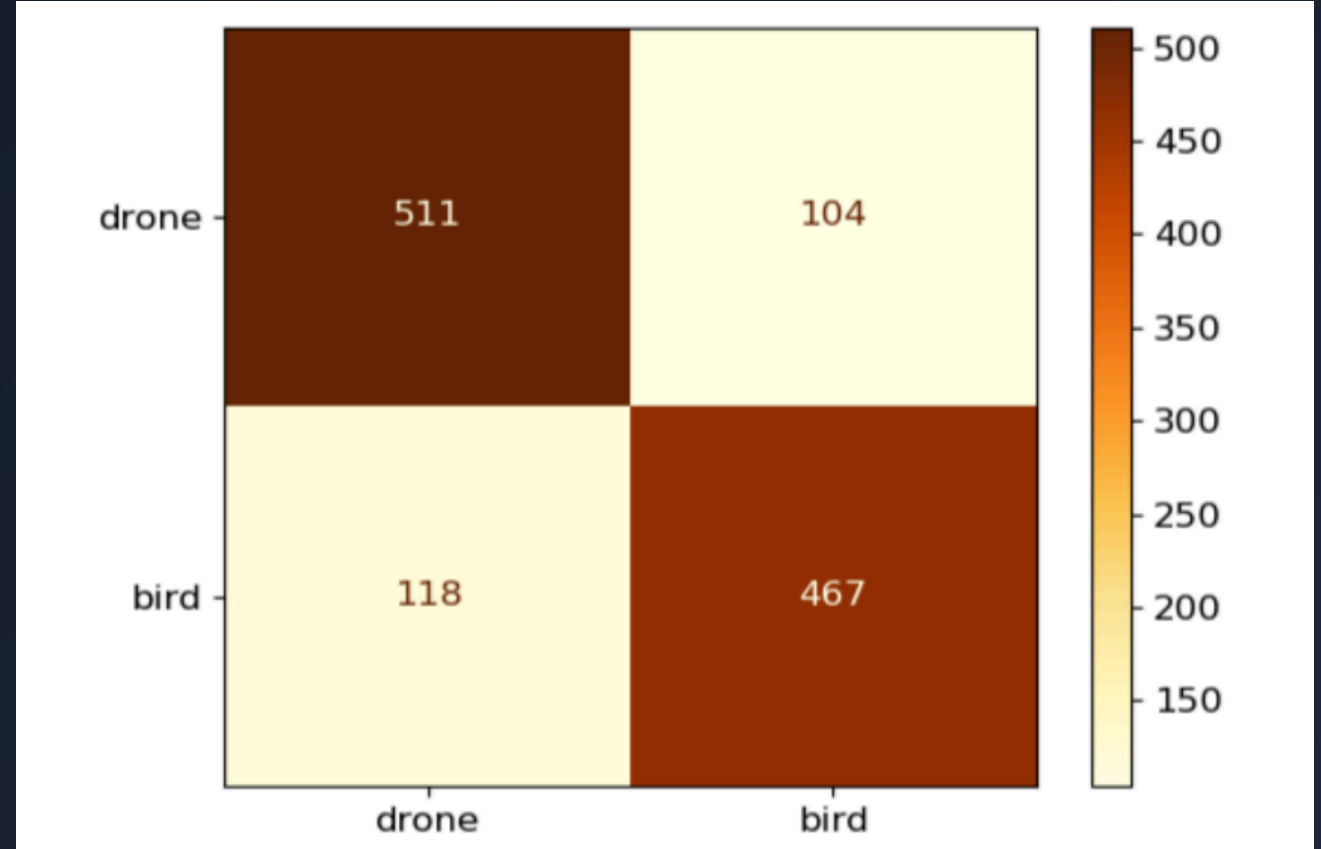


Fig. 3. Confusion matrix on the test dataset for the c1assification network.

Classification Accuracy vs. Distance

- Classification accuracy remains stable across distances
- Distance only slightly affects correct vs incorrect classification ratio
- Classifier generalizes well to longer distances due to sparse training samples

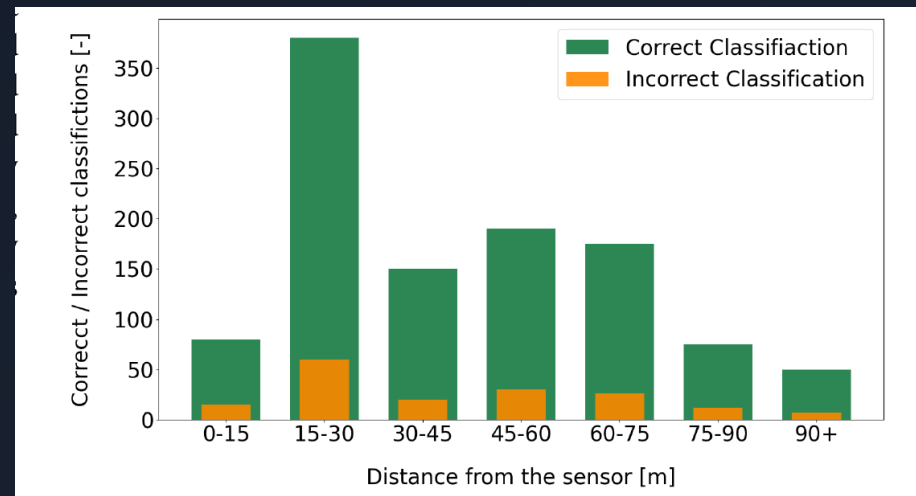


Fig. 4. The number of correct and incorrect classifications on the test dataset plotted against the object's distance from the sensor. As illustrated above, the distance only slightly affects the ratio of correct to incorrect classifications. This behaviour is the result of our collected samples being sparse to begin with, which greatly aided our classifier in generalizing for longer distances during training.

Conclusions and Future Work

- Proposed pipeline enables real-time UAV and bird detection using sparse LiDAR
- Attention-based segmentation and modified PointNet achieve robust performance
- Future work: Probabilistic analysis of detection limits and failure cases
- Further evaluation on optimal reflection thresholds and reliability at distance